

Who Issue Bonds?

Bonds are typically issued by corporations, governments (both national and local), and other organizations such as non-profits and universities.

When a corporation needs to raise funds for expansion or other purposes, they can issue bonds to investors in exchange for the promise to repay the principal amount plus interest at a future date. Similarly, governments issue bonds to fund public projects such as infrastructure development or to cover budget deficits.

Bonds can be bought and sold on the open market, and their value can fluctuate depending on a variety of factors, including interest rates, the creditworthiness of the issuer, and the maturity date of the bond.

Key Characteristics of Bonds

Bonds are a type of debt instrument used by corporations, governments, and other organizations to raise funds. The key characteristics of bonds include:

Face value: The face value of a bond is the amount that the issuer promises to pay the bondholder at the end of the bond's term (or maturity).

Coupon rate: The coupon rate is the interest rate that the issuer promises to pay the bondholder at regular intervals (usually annually or semi-annually) until the bond matures.

Maturity date: The maturity date is the date on which the issuer must repay the face value of the bond to the bondholder.

Yield to maturity: The yield to maturity is the total return that an investor can expect to earn on a bond if it is held until it matures.

Credit rating: The credit rating is an assessment of the creditworthiness of the issuer, and it can affect the interest rate that the issuer must pay to attract investors.

Liquidity: The liquidity of a bond refers to how easily it can be bought or sold on the open market. Bonds that are issued by large corporations or governments are typically more liquid than bonds issued by smaller organizations.

Call ability: Some bonds are "callable," which means that the issuer can choose to repay the bondholder before the maturity date. Callable bonds usually offer higher interest rates than non-callable bonds, but they carry more risk for the investor.

Bond Valuation

Bond valuation is the process of determining the fair price of a bond, which is the present value of its future cash flows. The following are the key steps involved in the bond valuation process:

Determine the coupon payments: The coupon payments are the regular interest payments that the bondholder will receive over the life of the bond. To determine the coupon payments, multiply the face value of the bond by the coupon rate.

Determine the required rate of return: The required rate of return is the rate of return that an investor requires to invest in the bond. This rate is based on factors such as the investor's risk tolerance, the current interest rate environment, and the creditworthiness of the issuer.

Calculate the present value of the coupon payments: The present value of the coupon payments is the sum of the discounted cash flows of the coupon payments. To calculate this, divide each coupon payment by $(1 + \text{the required rate of return})$ raised to the power of the number of periods until the payment is received, and then sum up these present values.

Calculate the present value of the face value: The present value of the face value is the discounted cash flow of the principal payment received at maturity. To calculate this, divide the face value by $(1 + \text{the required rate of return})$ raised to the power of the number of periods until the bond matures.

Add the present values of the coupon payments and the face value: The fair price of the bond is the sum of the present value of the coupon payments and the present value of the face value.

In summary, bond valuation is based on the time value of money principle, where future cash flows are discounted back to the present using an appropriate discount rate.

Bond Yields

Bond yield is a measure of the return an investor can expect to earn from a bond. The yield is typically expressed as a percentage of the bond's face value and can be calculated in several ways:

Coupon Yield: This is the annual interest rate on a bond, expressed as a percentage of the bond's face value. It is calculated by dividing the annual coupon payment by the face value of the bond.

Current Yield: This is the annual return an investor can expect to earn on a bond based on its current market price. It is calculated by dividing the annual coupon payment by the current market price of the bond.

Yield to Maturity (YTM): This is the total return an investor can expect to earn if the bond is held until maturity. It takes into account both the coupon payments and any capital gain or loss that will

result from the difference between the purchase price and the face value of the bond at maturity. YTM is calculated using a formula that considers the bond's current market price, the coupon rate, the time to maturity, and the face value.

Yield to Call (YTC): This is the yield an investor can expect to earn if the bond is called (redeemed) by the issuer before maturity. It is calculated using a formula that considers the call price, the coupon rate, the time to call, and the bond's current market price.

Yield to Worst (YTW): This is the lowest yield an investor can expect to earn based on the worst-case scenario for the bond, such as if it is called or defaults. It takes into account the yields to call and yield to maturity and considers which scenario results in the lowest yield.

The yield of a bond can be influenced by factors such as changes in interest rates, creditworthiness of the issuer, inflation expectations, and supply and demand dynamics in the bond market.

Changes in Bond Value over Time

The value of a bond can change over time due to a variety of factors, including changes in interest rates, creditworthiness of the issuer, and market conditions. Here are some examples of how changes in these factors can impact the value of a bond:

Interest Rates: When interest rates rise, the value of existing bonds decreases, because the fixed coupon payments of the bond become less attractive compared to new bonds that offer higher yields. Conversely, when interest rates fall, the value of existing bonds increases, because the fixed coupon payments become more attractive compared to new bonds that offer lower yields.

Creditworthiness: If the creditworthiness of the issuer declines, the value of its bonds will decrease, because investors will demand a higher return to compensate for the increased risk of default. On the other hand, if the creditworthiness of the issuer improves, the value of its bonds will increase, because investors will be willing to accept a lower return for the reduced risk of default.

Market Conditions: Supply and demand dynamics in the bond market can also impact the value of a bond. If there is high demand for a particular bond, its price will increase, and its yield will decrease. Conversely, if there is low demand for a bond, its price will decrease, and its yield will increase.

Overall, changes in interest rates have the most significant impact on the value of bonds, particularly for longer-term bonds, which are more sensitive to interest rate changes. However, changes in creditworthiness and market conditions can also affect bond values, and investors should consider these factors when evaluating the performance of their bond investments.

Bonds with Semiannual Coupons

Bonds with semiannual coupons are bonds that pay interest to investors twice a year, rather than once a year. These types of bonds are common in the bond market and are often referred to as semiannual bonds or semi-annual coupon bonds.

The semiannual coupon payment is calculated by dividing the annual coupon payment by two. For example, if a bond has a face value of \$1,000 and a coupon rate of 6%, the annual coupon payment would be \$60. The semiannual coupon payment would be \$30 ($\$60/2$).

Investors who hold bonds with semiannual coupons receive a payment every six months, which can provide a more consistent cash flow than bonds that pay interest annually. This can be particularly attractive for investors who rely on the income from their bond investments to meet their cash flow needs.

When valuing a bond with semiannual coupons, the coupon rate is typically divided by two to reflect the semiannual payment schedule. The other components of bond valuation, such as the required rate of return and the time to maturity, remain the same.

Overall, bonds with semiannual coupons are a popular choice for investors seeking regular income from their bond investments. They offer a steady stream of semiannual payments and can be an effective way to diversify a portfolio and manage risk.

Assessing Bond's Riskiness

Assessing the riskiness of a bond involves evaluating the likelihood that the issuer will default on the bond and the potential loss that investors may incur if the bond defaults. Here are some key factors to consider when assessing the risk of a bond:

Credit Rating: The credit rating of a bond is an evaluation of the issuer's creditworthiness, based on its financial strength and ability to meet its debt obligations. Bonds with higher credit ratings are generally considered less risky than bonds with lower credit ratings.

Yield: The yield of a bond reflects the compensation investors require for holding the bond and the perceived risk of default. Higher yields typically indicate higher perceived risk, while lower yields indicate lower perceived risk.

Maturity: Bonds with longer maturities are generally considered more risky than bonds with shorter maturities, because there is a greater chance that the issuer may default over a longer period of time.

Collateral: Bonds that are secured by collateral, such as assets or property, are generally considered less risky than unsecured bonds, because there is a greater chance that investors will be able to recover their investment in the event of a default.

Industry and Economic Conditions: The risk of a bond can also be influenced by external factors, such as changes in industry or economic conditions. For example, bonds issued by companies in industries that are experiencing significant disruption may be considered more risky than bonds issued by companies in stable industries.

Overall, assessing the riskiness of a bond requires a comprehensive evaluation of a range of factors, including credit rating, yield, maturity, collateral, and external conditions. Investors should carefully consider these factors when evaluating the risk and return potential of their bond investments.

Default Risk

Default risk is the risk that an issuer of a bond will fail to make interest payments or fail to repay the principal amount at maturity. This risk is particularly relevant for corporate bonds, where the issuer is a company rather than a government.

The likelihood of default depends on a number of factors, including the financial strength and creditworthiness of the issuer, industry and economic conditions, and the maturity of the bond. Bonds with higher credit ratings are generally considered less risky than bonds with lower credit ratings, as higher-rated issuers are perceived to be less likely to default.

To measure default risk, credit rating agencies such as Moody's, Standard & Poor's, and Fitch assign credit ratings to bond issuers based on their creditworthiness. These ratings range from AAA (highest credit quality) to D (default), with intermediate ratings such as A, B, and C indicating varying levels of credit risk.

Investors can also use other indicators of default risk, such as credit spreads, which reflect the additional yield that investors require for holding a bond compared to a risk-free asset like a government bond. Higher credit spreads indicate higher perceived default risk, while lower credit spreads indicate lower perceived default risk.

Overall, default risk is an important consideration for investors when evaluating bond investments. Investors should carefully consider the credit rating and other indicators of default risk when selecting bonds for their portfolios, and diversify their investments across a range of issuers and industries to manage risk.

Bond Markets

The bond market, also known as the fixed income market, is a financial market where investors can buy and sell debt securities such as bonds, notes, and bills. The bond market is a key component of the global financial system, providing a source of financing for governments, corporations, and other entities.

The bond market is typically divided into two main categories: the primary market and the secondary market.

The primary market is where new bond issues are sold to investors for the first time. In the primary market, issuers such as governments, corporations, and municipalities work with investment banks and underwriters to determine the terms and conditions of the bond issue, including the interest rate, maturity date, and payment schedule. Once the terms have been established, the bonds are offered for sale to investors in a process known as an initial public offering (IPO).

The secondary market is where previously issued bonds are bought and sold among investors. The secondary market is much larger than the primary market, and provides investors with the ability to buy and sell bonds after the initial offering. The secondary market is also where the price of a bond is determined, based on supply and demand factors, interest rates, and other market conditions.

Bond markets can also be classified by the type of issuer, such as government bonds, corporate bonds, municipal bonds, and international bonds. Each of these markets has its own unique characteristics and risks, and investors should carefully consider the specific market and issuer when selecting bonds for their portfolios.

Overall, the bond market plays a critical role in the global financial system, providing a source of financing for issuers and investment opportunities for investors. By offering a diverse range of bond options, the bond market can help investors achieve their financial goals while managing risk.